

EGR 3350-07

Technical Communications for Engineers and Computer Scientists

Ms. Alysoun Taylor-Hall

Mechanism Description Peer Review Draft



Wireless Fast Phone Charger

Utsav Acharya

Computer Science

February 5, 2026

Introduction

A wireless fast phone charger is a power-transfer device that supplies electrical energy to a smartphone without a direct wired connection. Energy transfers when a compatible smartphone is placed on the charger's upper surface. The charger described in this paper is circular, black, and constructed of rigid polymer plastic. It measures approximately 10 cm in diameter and 0.8 cm in height. The mechanism consists of four primary parts: the charging surface, the indicator light ring, the protective casing, and the power input port.

Discussion

Figure 1 shows the wireless fast phone charger and its primary components.

Charging Surface

The charging surface is a circular support platform that stabilizes a smartphone during electromagnetic energy transfer. It measures approximately 10 cm in diameter and 0.8 cm in thickness. The matte plastic surface has a slight texture to increase friction and prevent slipping. The flat circular design keeps the phone centered, allowing consistent electromagnetic induction.

Indicator Light Ring

The indicator light ring is a visual signaling component that communicates charging status. The ring follows the perimeter of the charging surface and has a thickness of approximately 0.2 cm. Constructed of translucent plastic, it emits colored LED light when connected to power. The light provides observable feedback regarding operational status.

Protective Casing

The protective casing is a structural enclosure that houses and supports the internal components. The casing measures approximately 9.5 cm in diameter and 0.8 cm in height. It is constructed of

rigid black polymer plastic with a smooth finish. The casing protects the internal induction coil and circuitry. Small rubber pads on the underside reduce slipping during operation.

Power Input Port

The power input port is an electrical interface that allows the charger to receive energy from an external source. Located on the side of the casing, it accommodates a USB-C connector and measures approximately 8 mm in width. The port is reinforced to withstand repeated cable insertion and removal and supplies electrical current to the internal induction system.

Conclusion

A wireless fast phone charger is a power-transfer device that supplies electrical energy to a smartphone without requiring a direct wired connection. The charging surface supports the phone, the indicator light ring provides feedback, the protective casing houses internal components, and the power input port receives external power. Together, these components form a compact mechanism that supports mobile computing devices.



Figure 1: Wireless Fast Phone Charger, showing its parts.